

Layer Stack Legend

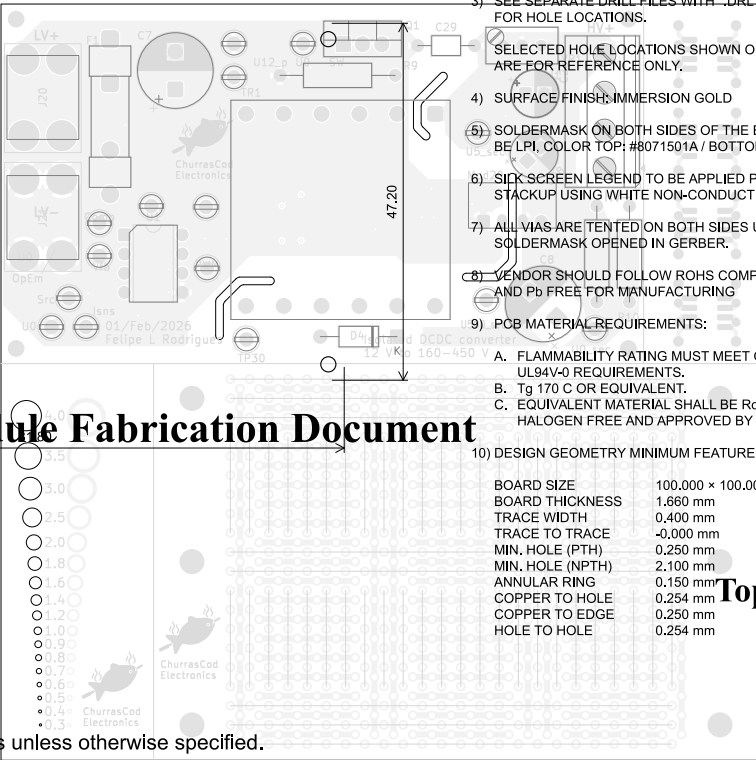
	Material	Layer	Thickness	Dielectric	Type	Gerber
	F.Paste				Paste Mask	
	F.Silkscreen				Legend	GBR
	F.Mask		0.02mm		Solder Resist	GBR
	Copper	L1 (Ref)	0.07mm (2.00oz)		Plane	GBR
	Prepreg		0.1mm	FR4_7628	Dielectric	
	Copper	Virtual (Top side jumpers)	0.035mm (1.00oz)		Copper Layer	GBR
	Core		1.21mm	FR4	Dielectric	
	Copper	Virtual (Bot side jumpers)	0.035mm (1.00oz)		Copper Layer	GBR
	Prepreg		0.1mm	FR4	Dielectric	
	Copper	L2 (Sig, PWR)	0.07mm (2.00oz)		Signal	GBR
	B.Mask		0.02mm		Solder Resist	GBR
	B.Silkscreen				Legend	GBR
	B.Paste				Paste Mask	

Total thickness: 1.66mm
Note: external layer thicknesses are specified after plating

Impedance Table

Transmission Line	Impedance [ohms]	Tolerance [ohms]	Layer	Trace Width [mm]	Gap [mm]	Ref. Layers
NA	-	-	-	-	-	-

Module Fabrication Document



FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

- FABRICATE PER IPC-6012A CLASS 2.
- OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge_Cuts,GBR" SUFFIX.

DIMENSIONS OF CIRCUMSIZED RECTANGLE SHOWN ON THIS DRAWING ARE FOR REFERENCE ONLY.
- SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.

SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING ARE FOR REFERENCE ONLY.
- SURFACE FINISH: IMMERSION GOLD
- SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR TOP: #8071501A / BOTTOM: GREEN.
- SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING WHITE NON-CONDUCTIVE EPOXY INK.
- ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDERMASK OPENED IN GERBER.
- VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING
- PCB MATERIAL REQUIREMENTS:

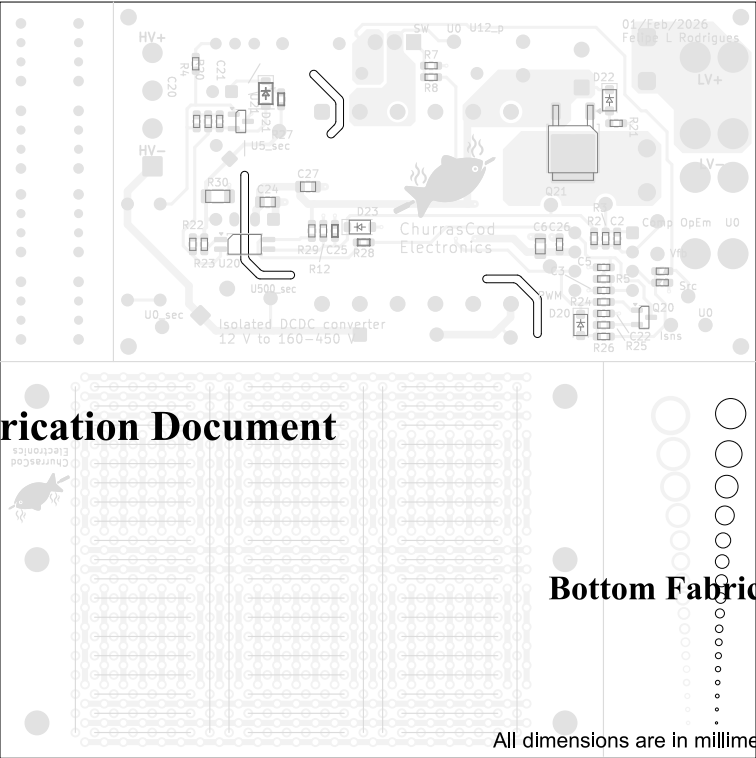
A. FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS.
B. Tg 170 C OR EQUIVALENT.
C. EQUIVALENT MATERIAL SHALL BE ROHS COMPLIANT, HALOGEN FREE AND APPROVED BY FR.
- DESIGN GEOMETRY MINIMUM FEATURE SIZES:

BOARD SIZE 100.000 x 100.000 mm
BOARD THICKNESS 1.660 mm
TRACE WIDTH 0.400 mm
TRACE TO TRACE -0.000 mm
MIN. HOLE (PTH) 0.250 mm
MIN. HOLE (NPTH) 2.100 mm
ANNULAR RING 0.150 mm
COPPER TO HOLE 0.254 mm
COPPER TO EDGE 0.250 mm
HOLE TO HOLE 0.254 mm

Top Fabrication (Scale 1:1)

All dimensions are in millimeters unless otherwise specified.

	Comments:	Company:		Variant:	Git Hash:
	Comment 1	FR		PRELIMINARY	af0b929
	Comment 2	Board Name:		Project Name:	
	Comment 3	Module		Isolated step-up 12:450 V	
	Sheet Title:	File Name:	Designer:	Date:	Revision:
	Top Fabrication (Scale 1:1)	Isolated_12to450V.kicad_pcb	FR	2026-02-02	1.0.0+ (Unreleased)
	Sheet Path:	Reviewer:		Size:	Sheet:
		NA		A3	1 of 10



Module Fabrication Document

Bottom Fabrication (Scale 1:1)

All dimensions are in millimeters unless otherwise specified.

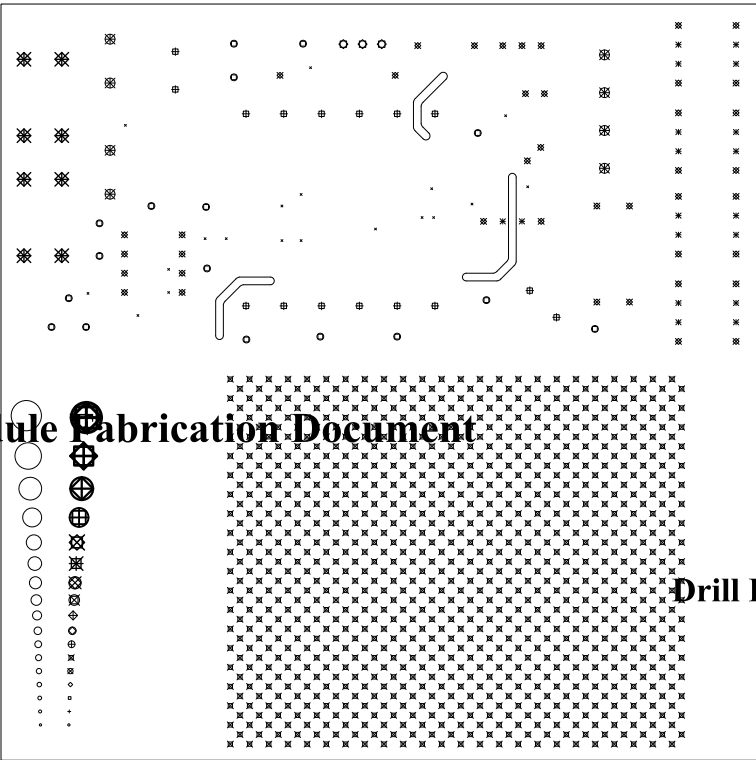
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	Comment 3				Isolated step-up 12:450 V	
	Comment 4					
	Sheet Title:	File Name:	Designer:	Date:	Revision:	
	Bottom Fabrication (Scale 1:1)	Isolated_12to450V.kicad_pcb	FR	2026-02-02	1.0.0+ (Unreleased)	
	Sheet Path:		Reviewer:	Size:	Sheet:	
		NA	A3	2 of 10		

Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	19	0,25mm (9,84mils)	PTH	Round	L1 (Ref) - L2 (Slg, PWR)	Via
○	1	0,30mm (11,81mils)	PTH	Round	L1 (Ref) - L2 (Slg, PWR)	Via
+	1	0,40mm (15,75mils)	PTH	Round	L1 (Ref) - L2 (Slg, PWR)	Via
□	1	0,50mm (19,69mils)	PTH	Round	L1 (Ref) - L2 (Slg, PWR)	Via
◇	1	0,60mm (23,62mils)	PTH	Round	L1 (Ref) - L2 (Slg, PWR)	Via
⊗	1	0,70mm (27,56mils)	PTH	Round	L1 (Ref) - L2 (Slg, PWR)	Via
※	18	0,70mm (27,56mils)	PTH	Round	L1 (Ref) - L2 (Slg, PWR)	Pad
※	937	0,80mm (31,50mils)	PTH	Round	L1 (Ref) - L2 (Slg, PWR)	Via
※	41	0,80mm (31,50mils)	PTH	Round	L1 (Ref) - L2 (Slg, PWR)	Pad
⊕	1	0,90mm (35,43mils)	PTH	Round	L1 (Ref) - L2 (Slg, PWR)	Via
○	17	0,90mm (35,43mils)	PTH	Round	L1 (Ref) - L2 (Slg, PWR)	Pad
○	1	1,00mm (39,37mils)	PTH	Round	L1 (Ref) - L2 (Slg, PWR)	Via
⊕	16	1,00mm (39,37mils)	PTH	Round	L1 (Ref) - L2 (Slg, PWR)	Pad
⊕	1	1,20mm (47,24mils)	PTH	Round	L1 (Ref) - L2 (Slg, PWR)	Via
○	3	1,20mm (47,24mils)	PTH	Round	L1 (Ref) - L2 (Slg, PWR)	Pad
※	8	1,30mm (51,18mils)	PTH	Round	L1 (Ref) - L2 (Slg, PWR)	Pad
※	1	1,40mm (55,12mils)	PTH	Round	L1 (Ref) - L2 (Slg, PWR)	Via
※	1	1,60mm (62,99mils)	PTH	Round	L1 (Ref) - L2 (Slg, PWR)	Via
※	1	1,80mm (70,87mils)	PTH	Round	L1 (Ref) - L2 (Slg, PWR)	Via
※	8	1,80mm (70,87mils)	PTH	Round	L1 (Ref) - L2 (Slg, PWR)	Pad
※	1	2,00mm (78,74mils)	PTH	Round	L1 (Ref) - L2 (Slg, PWR)	Via
⊕	1	2,50mm (98,43mils)	PTH	Round	L1 (Ref) - L2 (Slg, PWR)	Via
⊕	1	3,00mm (118,11mils)	PTH	Round	L1 (Ref) - L2 (Slg, PWR)	Via
⊕	1	3,50mm (137,80mils)	PTH	Round	L1 (Ref) - L2 (Slg, PWR)	Via
⊕	1	4,00mm (157,48mils)	PTH	Round	L1 (Ref) - L2 (Slg, PWR)	Via
Total 1083						

Module Fabrication Document

Drill Drawing L1 - L4 (Scale 1:1)



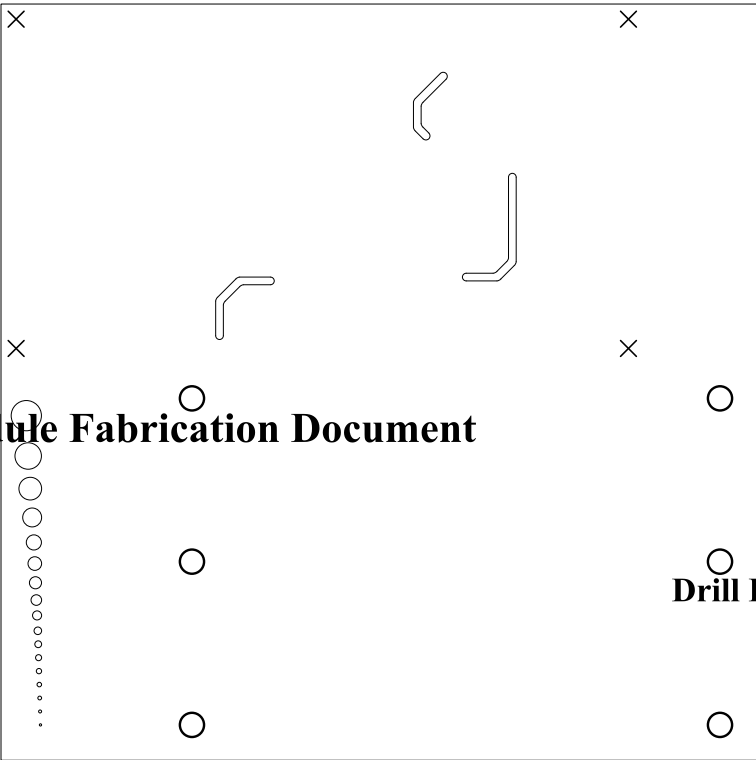
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			Board Name: Module		Project Name: Isolated step-up 12:450 V	
	Sheet Title: Drill Drawing (L1 - L4)		File Name: Isolated_12to450V.kicad_pcb	Designer: FR	Date: 2026-02-02	Revision: 1.0.0+ (Unreleased)
	Sheet Path:			Reviewer: NA	Size: A3	Sheet: 3 of 10

Drill Table

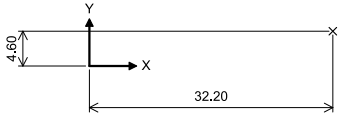
Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
X	4	2,10mm (82,68mils)	NPTH	Round	L1 (Ref) - L2 (Sig, PWR)	Mechanical
O	6	3,20mm (125,98mils)	NPTH	Round	L1 (Ref) - L2 (Sig, PWR)	Mechanical
	Total 10					

Module Fabrication Document

Drill Drawing L1 - L4 (Scale 1:1)

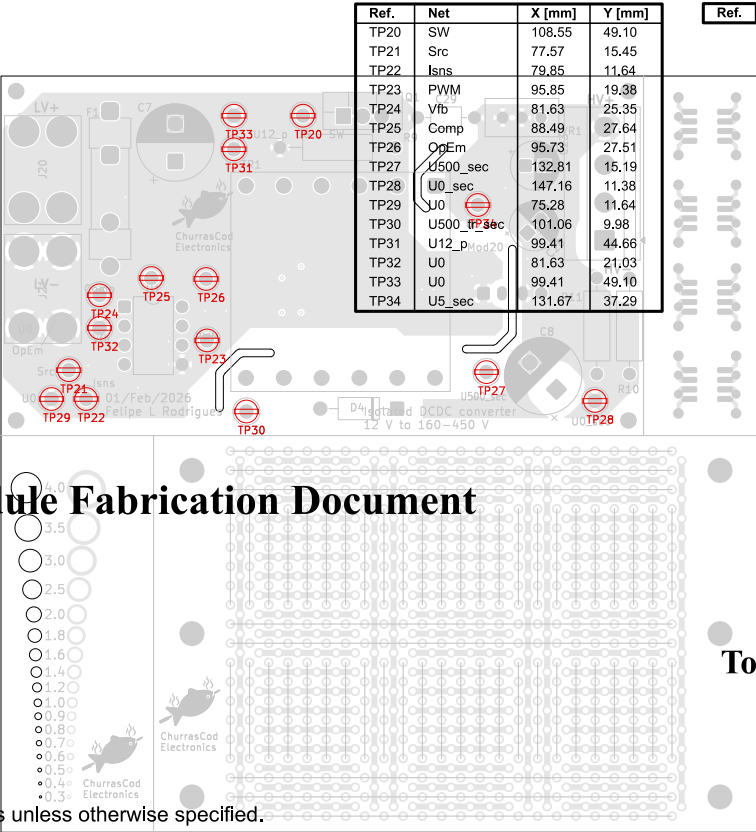


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			Board Name: Module		Project Name: Isolated step-up 12:450 V	
	Sheet Title: Drill Drawing (L1 - L4)		File Name: Isolated_12to450V.kicad_pcb	Designer: FR	Date: 2026-02-02	Revision: 1.0.0+ (Unreleased)
	Sheet Path:			Reviewer: NA	Size: A3	Sheet: 4 of 10



All dimensions are in millimeters unless otherwise specified.

Module Fabrication Document



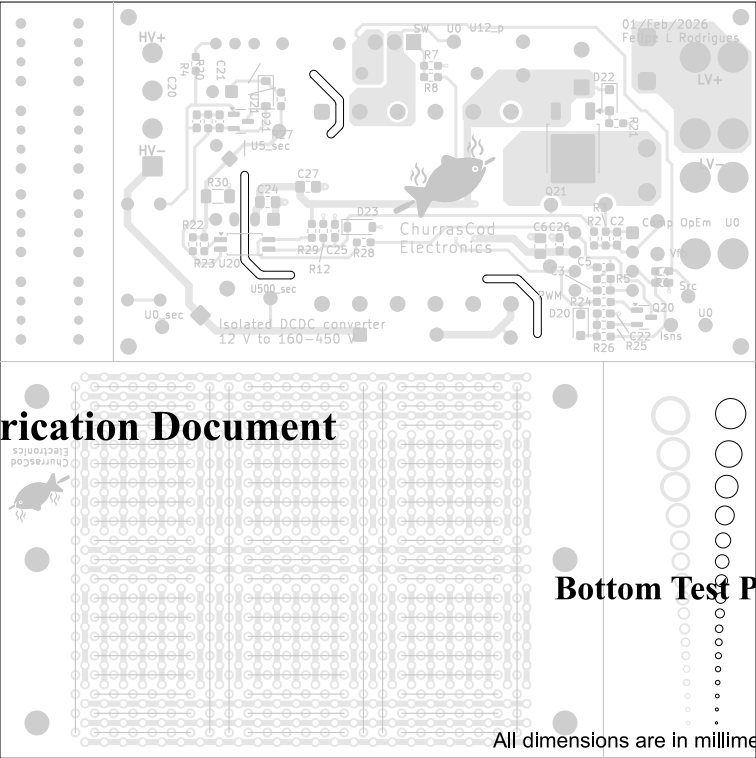
Ref.	Net	X [mm]	Y [mm]
TP20	SW	108.55	49.10
TP21	Src	77.57	15.45
TP22	Isns	79.85	11.64
TP23	PWM	95.85	19.38
TP24	Vfb	81.63	25.35
TP25	Comp	88.49	27.64
TP26	OpEm	95.73	27.51
TP27	U500_sec	132.81	15.19
TP28	U0_sec	147.16	11.38
TP29	U0	75.28	11.64
TP30	U500_sec	101.06	9.98
TP31	U12_PMod20	99.41	44.66
TP32	U0	81.63	21.03
TP33	U0	99.41	49.10
TP34	U5_sec	131.67	37.29

Ref.	Net	X [mm]	Y [mm]
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Top Test Points (Scale 1:1)

	Comments:	Company:		Variant:	Git Hash:
	Comment 1	FR		PRELIMINARY	af0b929
	Comment 2	Board Name:		Project Name:	
	Comment 3	Module		Isolated step-up 12:450 V	
	Comment 4	File Name:		Designer:	Date:
Sheet Title:		Top Test Points (Scale 1:1)		FR	2026-02-02
Sheet Path:		Isolated_12to450V.kicad_pcb		Revision:	1.0.0+ (Unreleased)
		Reviewer:		Size:	Sheet:
		NA		A3	5 of 10

Ref.	Net	X [mm]	Y [mm]
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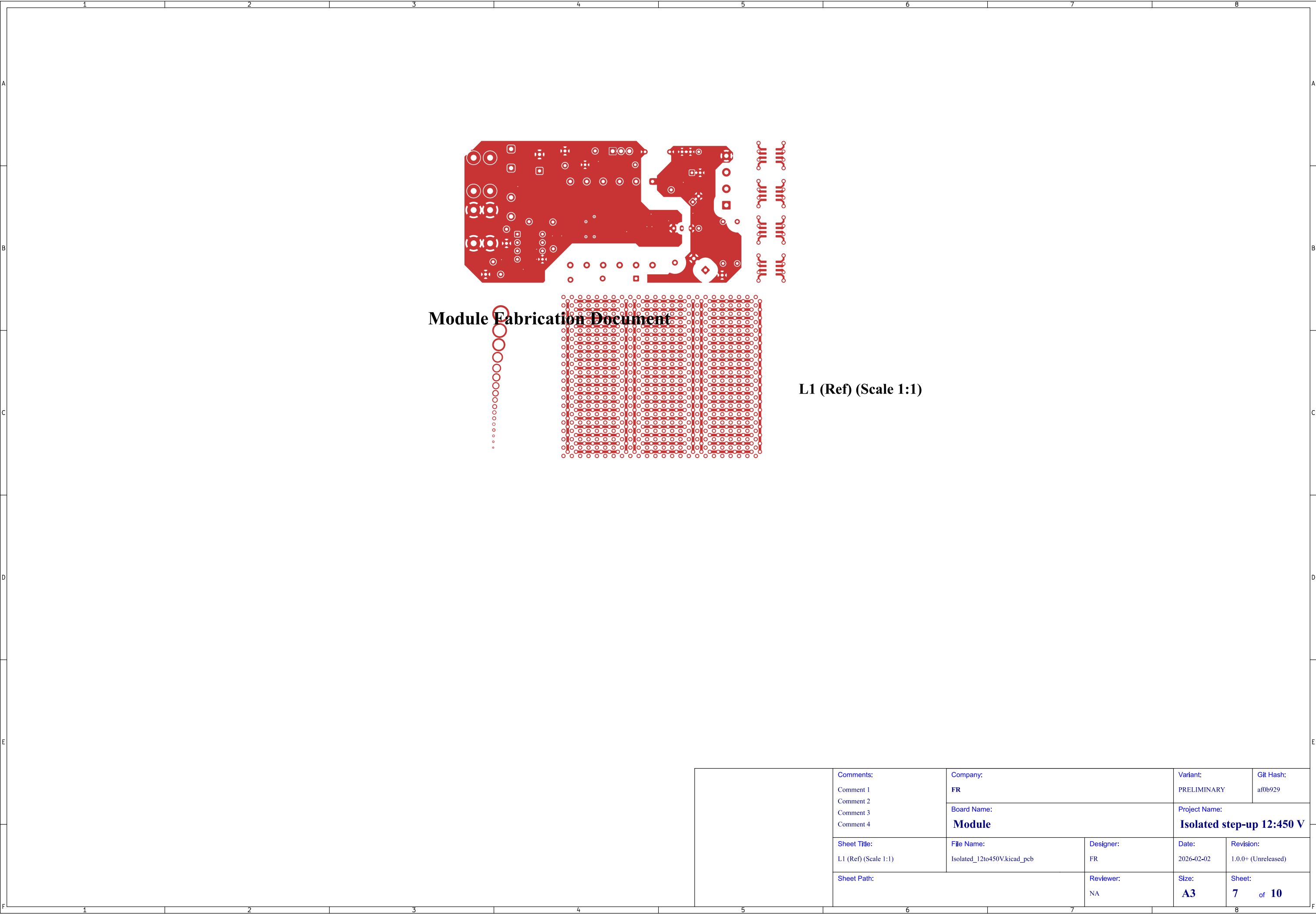


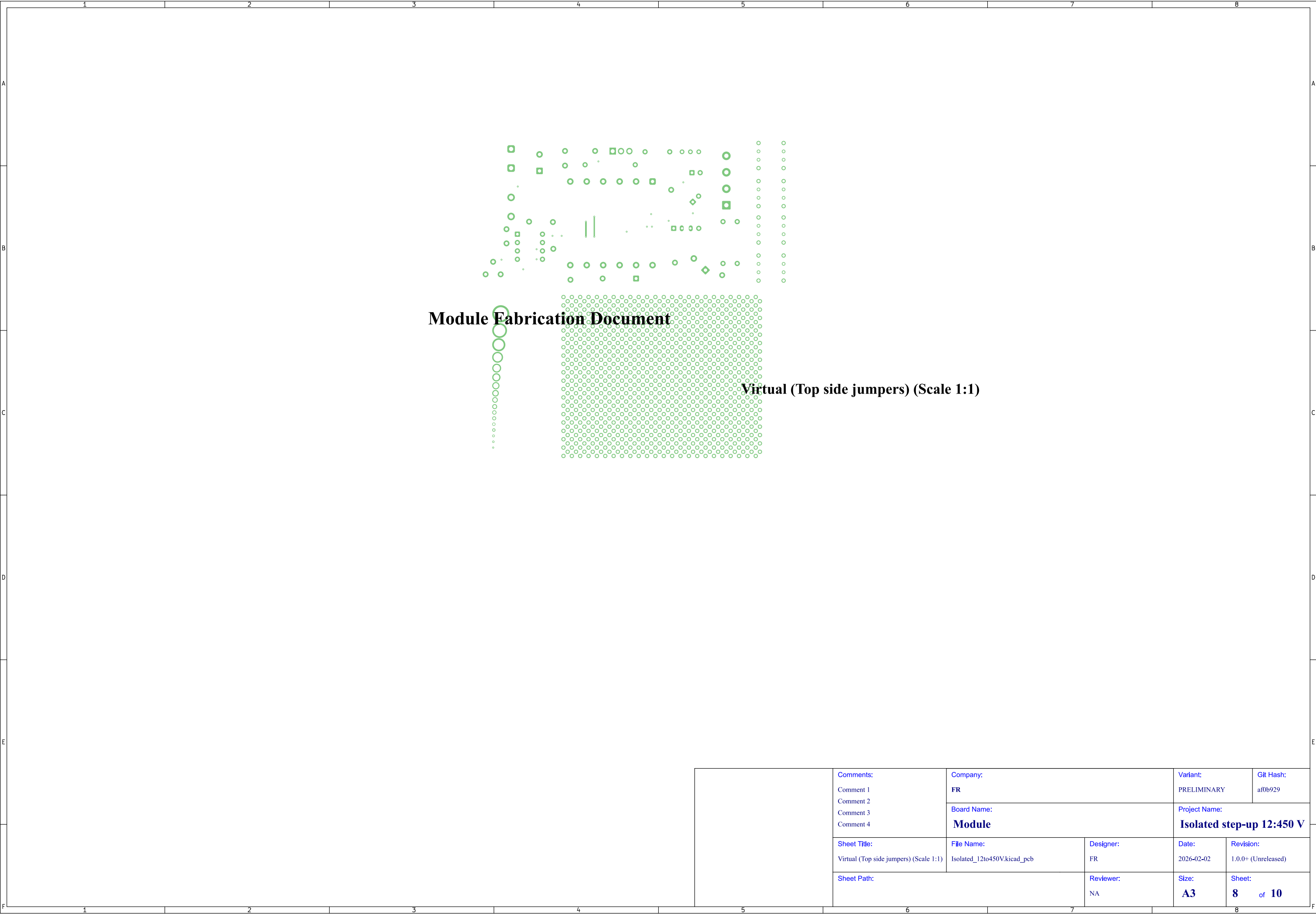
Module Fabrication Document

Bottom Test Points (Scale 1:1)

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	Comment 1	FR		PRELIMINARY	af0b929
	Comment 2	Board Name:		Project Name:	
	Comment 3	Module		Isolated step-up 12:450 V	
	Comment 4	File Name:		Designer:	Date:
		Isolated_12to450V.kicad_pcb		FR	2026-02-02
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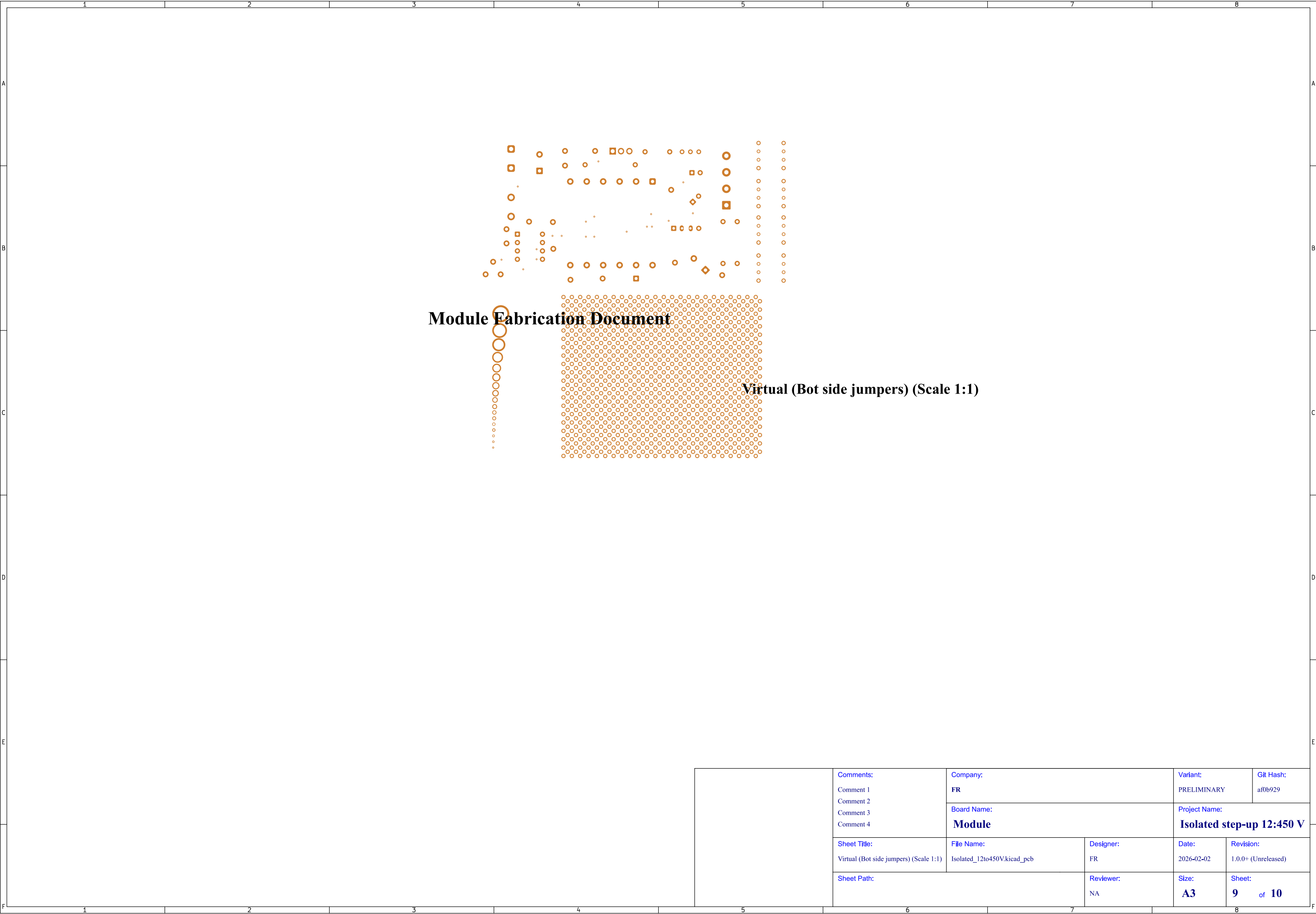




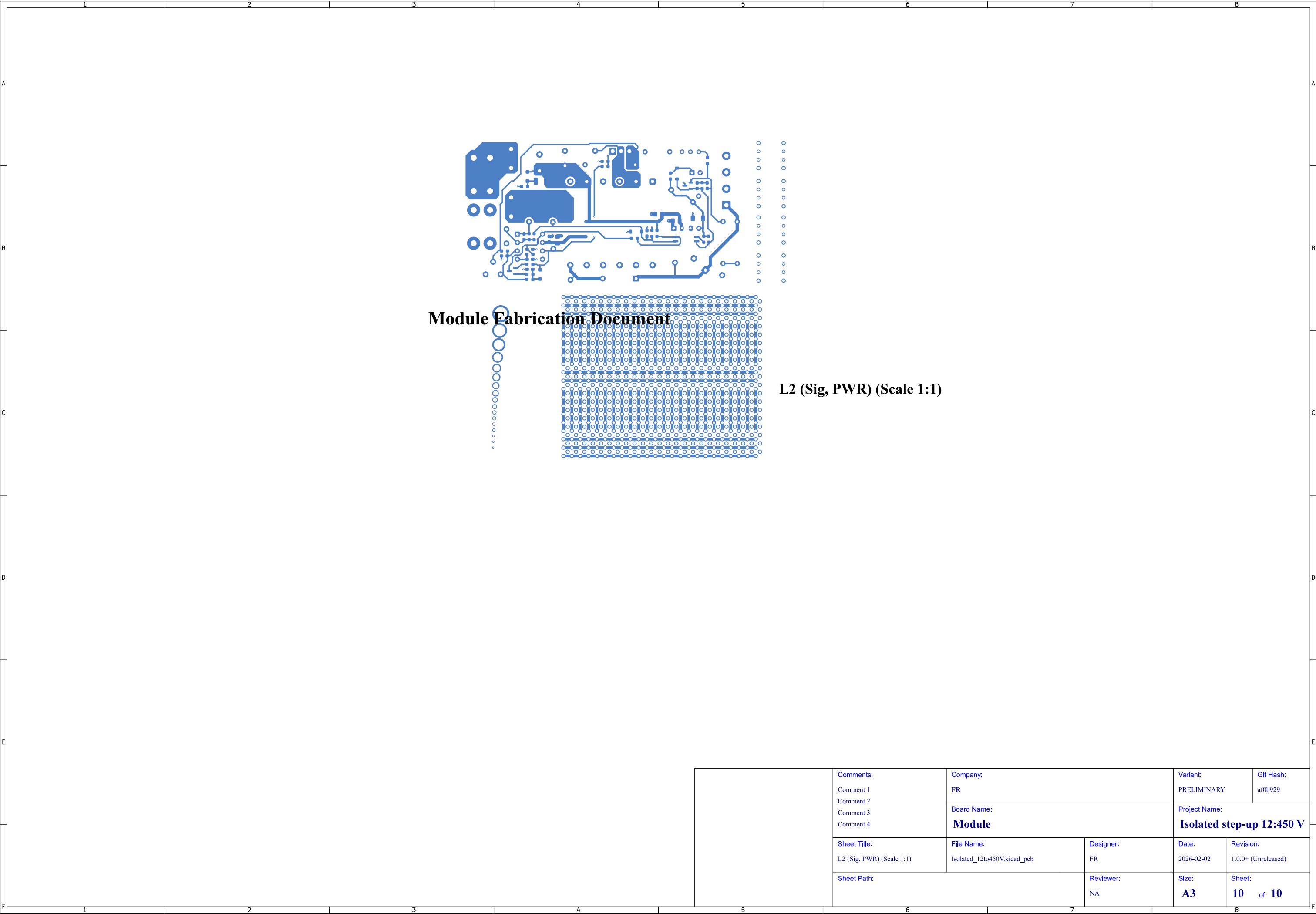
Module Fabrication Document

Virtual (Top side jumpers) (Scale 1:1)

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			Board Name: Module		Project Name: Isolated step-up 12:450 V	
	Sheet Title: Virtual (Top side jumpers) (Scale 1:1)		File Name: Isolated_12to450V.kicad_pcb	Designer: FR	Date: 2026-02-02	Revision: 1.0.0+ (Unreleased)
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	Comments: Comment 1 Comment 2 Comment 3 Comment 4		Company: FR		Variant: PRELIMINARY	Git Hash: af0b929
			Board Name: Module		Project Name: Isolated step-up 12:450 V	
	Sheet Title: Virtual (Bot side jumpers) (Scale 1:1)		File Name: Isolated_12to450V.kicad_pcb	Designer: FR	Date: 2026-02-02	Revision: 1.0.0+ (Unreleased)
	Sheet Path:			Reviewer: NA	Size: A3	Sheet: 9 of 10



Module Fabrication Document

L2 (Sig, PWR) (Scale 1:1)

	Comments:	Company:		Variant:		Git Hash:		
	Comment 1	FR		PRELIMINARY		af0b929		
	Comment 2	Board Name: Module			Project Name: Isolated step-up 12:450 V			
	Comment 3							
	Comment 4							
	Sheet Title:	File Name:		Designer:		Date:	Revision:	
	L2 (Sig. PWR) (Scale 1:1)	Isolated_12to450V.kicad_pcb		FR		2026-02-02	1.0.0+ (Unreleased)	
Sheet Path:			Reviewer:		Size:	Sheet:		
			NA		A3	10 of 10		